FROM REAL TO COMPLEX MATRIX CONDITIONING

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1. Starting Theorem

We begin by stating the following result from "Complex Matrix Inversion via Real Matrix Inversions"[1] which was the motivation for this report:

Corollary 1.1. *Let $A \in GL_n(\mathbb{R})$ and $B \in \mathbb{R}^{n\times n}$. Let $A = QR$ be a QR decomposition with $Q \in O_n(\mathbb{R})$ and $R \in \mathbb{R}^{n\times n}$ upper triangular. If $N = Q^{\mathsf{T}}BR^{-1}$ is a normal matrix with eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, then*

$$\kappa(A)\frac{\max_{k=1,\ldots,n}|1+i\lambda_k|}{\min_{k=1,\ldots,n}|1+i\lambda_k|} \geq \kappa(A+iB) \geq \max_{k=1,\ldots,n}\frac{1}{|1+i\lambda_k|}.$$

We examine the two families of B to construct every normal N in $\mathbb{R}^{2\times 2}$. Specifically, when can we use Corollary 3.2? Given a 2×2 real invertible matrix A and after finding its QR decomposition, when is $N = Q^{-1}BR^{-1} = Q^TBR^{-1}$ normal (given $B \in \mathbb{R}^{2\times 2}$)?

First of all, since $A \in GL_2(\mathbb{R})$, it has a QR decomposition. The particular QR factorization used below is fixed by requiring the first column of Q to be the normalized first column of A and $\det(Q) = 1$. This convention determines the QR factorization uniquely.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix},$$

where

$$\det(A) = ad - bc \neq 0.$$

The orthogonal matrix Q and the upper triangular matrix R are given by

$$Q = \frac{1}{\sqrt{a^2+c^2}}\begin{pmatrix} a & -c \\ c & a \end{pmatrix},$$

and

$$R = Q^TA = \frac{1}{\sqrt{a^2+c^2}}\begin{pmatrix} a^2+c^2 & ab+cd \\ 0 & ad-bc \end{pmatrix}.$$

Here $Q^T = Q^{-1}$.

We evaluate

$$C = A + iB = QR + iB = Q(R + iQ^TB).$$

By defining

$$N = Q^TBR^{-1},$$

we obtain

$$C = Q(I + iN)R.$$

Since Q is orthogonal, it preserves lengths. Therefore,

$$\kappa_2(C) = \kappa_2((I + iN)R).$$

For

$$s = \sqrt{a^2+c^2},$$

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and

$$t = ab + cd, \qquad \Delta = ad - bc,$$

we have

$$R^{-1} = \begin{pmatrix} 1/s & -t/(s\Delta) \\ 0 & s/\Delta \end{pmatrix}.$$

We compute

$$Q^T B = \frac{1}{s}\begin{pmatrix} a & c \\ -c & a \end{pmatrix}\begin{pmatrix} e & f \\ g & h \end{pmatrix} = \frac{1}{s}\begin{pmatrix} ae + cg & af + ch \\ -ce + ag & -cf + ah \end{pmatrix}.$$

This leads to

$$N = Q^T B R^{-1} :$$

$$\begin{aligned} N &= \frac{1}{s}\begin{pmatrix} ae + cg & af + ch \\ -ce + ag & -cf + ah \end{pmatrix}\begin{pmatrix} 1/s & -t/(s\Delta) \\ 0 & s/\Delta \end{pmatrix} \\ &= \begin{pmatrix} \frac{p}{s^2} & -\frac{pt}{s^2\Delta} + \frac{q}{\Delta} \\ \frac{r}{s^2} & -\frac{rt}{s^2\Delta} + \frac{u}{\Delta} \end{pmatrix}, \end{aligned}$$

where

$$p = ae + cg, \quad q = af + ch, \quad r = -ce + ag, \quad u = -cf + ah.$$

To determine when N is normal, we require

$$NN^T = N^T N.$$

Let

$$N = \begin{pmatrix} x & y \\ z & w \end{pmatrix}, \qquad N^T = \begin{pmatrix} x & z \\ y & w \end{pmatrix}.$$

Then

$$\begin{aligned} NN^T &= \begin{pmatrix} x & y \\ z & w \end{pmatrix}\begin{pmatrix} x & z \\ y & w \end{pmatrix} \\ &= \begin{pmatrix} x^2 + y^2 & xz + yw \\ xz + wy & z^2 + w^2 \end{pmatrix}, \end{aligned}$$

and

$$\begin{aligned} N^T N &= \begin{pmatrix} x & z \\ y & w \end{pmatrix}\begin{pmatrix} x & y \\ z & w \end{pmatrix} \\ &= \begin{pmatrix} x^2 + z^2 & xy + zw \\ xy + zw & y^2 + w^2 \end{pmatrix}. \end{aligned}$$

Equating the corresponding elements

- $$x^2 + y^2 = x^2 + z^2 \implies y^2 = z^2 \implies y = \pm z.$$
- $$xz + yw = xy + zw \implies (x - w)(y - z) = 0.$$

Thus, one of the following two cases must hold:

Case 1: $y = z$.
Case 2: $y = -z$ and $x = w$.

The two cases overlap when $y = z = 0$ and $x = w$.

1.1. **Case 1**: N **is Symmetric.** For Case 1, we have

$$\frac{r}{s^2} = -\frac{pt}{s^2\Delta} + \frac{q}{\Delta} \implies r\Delta = -pt + s^2 q.$$

For Case 2, the conditions are

$$\frac{r}{s^2} = \frac{pt}{s^2\Delta} - \frac{q}{\Delta} \implies r\Delta = pt - s^2 q$$

and

$$\frac{p}{s^2} = -\frac{rt}{s^2\Delta} + \frac{u}{\Delta} \implies p\Delta = -rt + us^2.$$

Expanding Case 1 gives

$$(ad - bc)(-ce + ag) = -(ae + cg)(ab + cd) + (af + ch)(a^2 + c^2).$$

After simplification, we obtain

$$af + ch = be + dg.$$

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix},$$

we have

$$A^T B = \begin{pmatrix} a & c \\ b & d \end{pmatrix}\begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae + cg & af + ch \\ be + dg & bf + dh \end{pmatrix},$$

and

$$(A^T B)^T = B^T A = \begin{pmatrix} ae + cg & be + dg \\ af + ch & bf + dh \end{pmatrix}.$$

Therefore, in Case 1, N is normal if and only if $A^T B$ is symmetric.

Even better, set

$$Y = BA^{-1} \implies B = YA.$$

Thus

$$A^T B = A^T Y A.$$

Since A is invertible,

$$\begin{aligned} (A^T B)^T = A^T B &\iff (A^T Y A)^T = A^T Y A \\ &\iff A^T Y^T A = A^T Y A \\ &\iff A^T (Y^T - Y) A = 0 \\ &\iff Y^T = Y. \end{aligned}$$

We have therefore shown that

$$A^T B \text{ symmetric} \iff BA^{-1} \text{ symmetric} \iff N \text{ symmetric}.$$

So $B = YA$ and B can be obtained by multiplying A by a symmetric matrix on the left.

This is a way to generate B for our results in Case 1: for any fixed invertible matrix $A \in \mathbb{R}^{2\times 2}$, every matrix B satisfying Case 1 has the form

$$B = \begin{pmatrix} p & q \\ q & r \end{pmatrix} A, \qquad p, q, r \in \mathbb{R}.$$

This can also be derived directly from the condition

$$y = z \implies N = \begin{pmatrix} x & y \\ y & w \end{pmatrix}.$$

Since $B = SA$ for a symmetric matrix S, we get

$$B = SQR.$$

Consequently,

$$N = Q^T BR^{-1} = Q^T SQRR^{-1} = Q^T SQ.$$

Taking the transpose,

$$N^T = Q^T S^T Q.$$

But $S = S^T$, so

$$N^T = Q^T SQ = N.$$

Because S can be any symmetric 2×2 matrix and A can be any invertible matrix, this construction can produce every symmetric N.

In this case,

$$C = A + iB = Q(I + iN)R.$$

1.2. **Case 2**: N **is a Scaled Rotation.** Here

$$N = \begin{pmatrix} x & y \\ -y & x \end{pmatrix} = xI + yJ,$$

where

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad J^2 = -I.$$

Then

$$\begin{aligned} B &= QNR \\ &= Q(xI + yJ)R \\ &= (xI + yQJQ^T)A. \end{aligned}$$

For any invertible A, we can choose any real x, y so that we construct B for N to be normal with

$$B = (xI + yQJQ^T)A,$$

similar to $B = SA$ in Case 1.

In Case 2, we construct a family of matrices that are not necessarily symmetric.

The two cases produce every normal matrix in $\mathbb{R}^{2\times 2}$ because every real normal matrix of this size is either symmetric or of the form

$$N = \begin{pmatrix} x & y \\ -y & x \end{pmatrix}.$$

Therefore, all normal matrices $N \in \mathbb{R}^{2\times 2}$ are attainable for any fixed invertible A by setting

$$B = SA$$

or

$$B = (xI + yQJQ^T)A.$$

- **In Case 1**:

$$C_1 = A + iB = A + iSA = (I + iS)A.$$

- **In Case 2**:

$$C_2 = Q(I + iN)R = Q\left(I + i\begin{pmatrix} x & y \\ -y & x \end{pmatrix}\right)R.$$

We can investigate how $\kappa(C)$ is affected by the entries of S in the first case and by x, y in the second, for a fixed A.

We could start with

$$A = \begin{pmatrix} 1 & 0 \\ 0 & \epsilon \end{pmatrix}$$

to test $\kappa(A)$ and compare it to $\kappa(C)$, and then consider general diagonal matrices

$$A = \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix}.$$

We care for N being normal since we can use the property that normal matrices are unitarily diagonalizable:

$$N = U\Lambda U^H,$$

where

$$UU^H = U^H U = I$$

and

$$\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

This will be useful for our proofs later.

Proof. For an eigenvalue λ of N, let

$$Nv = \lambda v.$$

Since $N = U\Lambda U^*$, we have

$$U\Lambda U^* v = \lambda v.$$

Multiplying by U^* gives

$$\Lambda U^* v = \lambda U^* v.$$

Let

$$w = U^* v.$$

Then

$$\Lambda w = \lambda w.$$

Since U is unitary and $v \neq 0$, we have $w \neq 0$. Therefore, λ is an eigenvalue of Λ.

Conversely, since N and Λ are unitarily similar, they have the same characteristic polynomial and hence the same eigenvalues. □

For example, for

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 10 \end{pmatrix}, \qquad S = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix},$$

we have $Q = I$, $R = A$, and

$$B = QSR = \begin{pmatrix} 2 & 0 \\ 0 & 10 \end{pmatrix}$$

(Case 1).

Then

$$C = (I + iS)A = \begin{pmatrix} 1 + 2i & 0 \\ 0 & 10(1 + i) \end{pmatrix}.$$

We see that

$$\kappa_2(A) = 10$$

and

$$\kappa_2(C) = \frac{10\sqrt{2}}{\sqrt{5}} = 2\sqrt{10} \simeq 6.325.$$

Thus

$$\kappa_2(A) > \kappa_2(C).$$

This is an example of achieving better conditioning on C than the one A has.

2. Conditioning Analysis

Lemma 2.1. *For*

$$A = \begin{pmatrix} 1 & 0 \\ 0 & \epsilon \end{pmatrix}, \qquad 0 < \epsilon < 1,$$

and

$$B = (xI + yQJQ^T)A,$$

the scaled-rotation case cannot improve the conditioning of

$$C = A + iB.$$

That is,

$$\kappa_2(C) \geq \kappa_2(A).$$

Proof. For this diagonal matrix, we may take $Q = I$ and $R = A$. Therefore,

$$B = NA = \begin{pmatrix} x & y \\ -y & x \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & \epsilon \end{pmatrix} = \begin{pmatrix} x & \epsilon y \\ -y & \epsilon x \end{pmatrix}.$$

We have

$$C = A + iB = \begin{pmatrix} 1 + ix & i\epsilon y \\ -iy & \epsilon(1 + ix) \end{pmatrix}.$$

Then

$$C^*C = \begin{pmatrix} 1 + x^2 + y^2 & 2i\epsilon y \\ -2i\epsilon y & \epsilon^2(1 + x^2 + y^2) \end{pmatrix}.$$

Let

$$s = 1 + x^2 + y^2.$$

The eigenvalues of C^*C are

$$\boxed{\lambda_\pm = \frac{s(1+\epsilon^2) \pm \sqrt{s^2(1-\epsilon^2)^2 + 16\epsilon^2 y^2}}{2}.}$$

Thus

$$\kappa_2(C) = \sqrt{\frac{\lambda_+}{\lambda_-}}$$

whenever C is invertible.

We know that

$$\kappa_2(C) \geq \frac{\max\{\|c_1\|_2, \|c_2\|_2\}}{\min\{\|c_1\|_2, \|c_2\|_2\}},$$

where c_1, c_2 are the columns of C.

The column norms are

$$\|c_1\|_2 = \sqrt{1 + x^2 + y^2}$$

and

$$\|c_2\|_2 = \epsilon\sqrt{1 + x^2 + y^2} = \epsilon\|c_1\|_2.$$

Since $0 < \epsilon < 1$, we have

$$\|c_1\|_2 > \|c_2\|_2.$$

Therefore,

$$\kappa_2(C) \geq \frac{\|c_1\|_2}{\|c_2\|_2} = \frac{1}{\epsilon}.$$

But

$$\kappa_2(A) = \frac{1}{\epsilon}.$$

Hence

$$\boxed{\kappa_2(C) \geq \kappa_2(A).}$$

If C is singular, its condition number is infinite and the conclusion is immediate. □

Lemma 2.2. *Let $C \in \mathbb{C}^{2\times 2}$ be invertible, with columns c_1, c_2. Then*

$$\kappa_2(C) \geq \frac{\max\{\|c_1\|_2, \|c_2\|_2\}}{\min\{\|c_1\|_2, \|c_2\|_2\}}.$$

Proof. By definition,

$$\kappa_2(C) = \|C\|_2 \|C^{-1}\|_2.$$

Since

$$\|C\|_2 = \max_{x\neq 0} \frac{\|Cx\|_2}{\|x\|_2},$$

we have

$$\|C\|_2 \geq \|Ce_1\|_2 = \|c_1\|_2$$

and similarly

$$\|C\|_2 \geq \|Ce_2\|_2 = \|c_2\|_2.$$

Thus

$$\|C\|_2 \geq \max\{\|c_1\|_2, \|c_2\|_2\}.$$

Also,

$$C^{-1}c_1 = e_1, \qquad C^{-1}c_2 = e_2.$$

Therefore,

$$\|C^{-1}\|_2 \geq \frac{1}{\|c_1\|_2}$$

and

$$\|C^{-1}\|_2 \geq \frac{1}{\|c_2\|_2}.$$

Hence

$$\|C^{-1}\|_2 \geq \frac{1}{\min\{\|c_1\|_2, \|c_2\|_2\}}.$$

Multiplying the two inequalities gives

$$\boxed{\kappa_2(C) \geq \frac{\max\{\|c_1\|_2, \|c_2\|_2\}}{\min\{\|c_1\|_2, \|c_2\|_2\}}.}$$

□

Lemma 2.3. *For any*

$$A = \operatorname{diag}(a, b), \qquad a, b > 0,$$

and

$$N = \begin{pmatrix} x & y \\ -y & x \end{pmatrix},$$

where $Q = I$ and $R = A$, we have

$$\kappa_2(C) \geq \kappa_2(A).$$

Proof. We have

$$B = NA = \begin{pmatrix} xa & yb \\ -ya & xb \end{pmatrix},$$

leading to

$$C = \begin{pmatrix} a(1+ix) & iyb \\ -iya & b(1+ix) \end{pmatrix}.$$

The columns are

$$c_1 = \begin{pmatrix} a(1+ix) \\ -iya \end{pmatrix}, \qquad c_2 = \begin{pmatrix} iyb \\ b(1+ix) \end{pmatrix}.$$

Their norms are

$$\|c_1\|_2 = a\sqrt{1+x^2+y^2}$$

and

$$\|c_2\|_2 = b\sqrt{1+x^2+y^2}.$$

Using the previous result,

$$\kappa_2(C) \geq \frac{\max\{a,b\}}{\min\{a,b\}}.$$

Since

$$\kappa_2(A) = \frac{\max\{a,b\}}{\min\{a,b\}},$$

we conclude that

$$\boxed{\kappa_2(C) \geq \kappa_2(A).}$$

□

Lemma 2.4. *The scaled-rotation obstruction is invariant under orthogonal changes of QR coordinates. In particular, for any positive diagonal matrix*

$$A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix},$$

the scaled-rotation construction cannot improve its condition number.

Proof. Let

$$N = \begin{pmatrix} x & y \\ -y & x \end{pmatrix}, \qquad B = QNR, \qquad C = A + iB,$$

where $A = QR$.

Since $R = Q^T A$, we have

$$C = (I + iQNQ^T)A.$$

We know that

$$QJQ^T = \det(Q)J$$

for every real orthogonal 2×2 matrix Q.

Indeed, for

$$Q = \begin{pmatrix} p & r \\ q & s \end{pmatrix},$$

direct multiplication gives

$$QJQ^T = \begin{pmatrix} 0 & ps - rq \\ rq - ps & 0 \end{pmatrix} = \det(Q)J.$$

Thus

$$QNQ^T = xI + y\det(Q)J.$$

Set

$$t = y\det(Q) \in \mathbb{R}.$$

Then

$$C = (I + i(xI + tJ))A.$$

For diagonal A, this gives

$$C = \begin{pmatrix} a(1+ix) & itb \\ -ita & b(1+ix) \end{pmatrix}.$$

The column norms are

$$\|c_1\|_2 = a\sqrt{1+x^2+t^2}$$

and

$$\|c_2\|_2 = b\sqrt{1+x^2+t^2}.$$

By the column-norm inequality,

$$\kappa_2(C) \geq \frac{\max\{a,b\}}{\min\{a,b\}} = \kappa_2(A).$$

This proves the result. □

As we have seen with the trivial cases above, case 2 does not look promising. We will check case 1 next.

2.1. **Finding Perfectly Conditioned C.** For

$$A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

and

$$N = \begin{pmatrix} x & y \\ y & z \end{pmatrix},$$

we try to construct a perfectly conditioned C using $B = QNR$.

Say

$$0 < a < b$$

without loss of generality.

We could use

$$\kappa_2(A) = \frac{\max(|a|,|b|)}{\min(|a|,|b|)},$$

but it would mean more calculations. Here,

$$\kappa_2(A) = \frac{b}{a}.$$

For $B = SA$, where

$$S = QNQ^T,$$

we know $R = Q^TA$ implies

$$B = QNR = QNQ^TA = SA.$$

Since S is symmetric, N is symmetric as shown earlier.

Let

$$B = \begin{pmatrix} u & v \\ v & w \end{pmatrix}\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} = \begin{pmatrix} ua & vb \\ va & wb \end{pmatrix}.$$

This gives

$$C = A + iB = \begin{pmatrix} a(1+iu) & ivb \\ iva & b(1+iw) \end{pmatrix}.$$

Can some u, v, w give us

$$\kappa_2(C) = 1?$$

We must prove that

$$\kappa_2(C) = 1 \iff C^*C = \lambda I$$

for some $\lambda > 0$, provided C is invertible.

Proof. We have

$$\kappa_2(C) = \frac{\sigma_{\max}(C)}{\sigma_{\min}(C)} = 1$$

if and only if all singular values of C are equal.

Say they are all equal to $s > 0$.

Since C^*C is a Hermitian positive semidefinite matrix, its eigenvalues are the squared singular values of C. Thus all its eigenvalues are equal to s^2.

By the spectral theorem,

$$C^*C = U(s^2I)U^* = s^2I.$$

Setting $\lambda = s^2$, we obtain

$$C^*C = \lambda I.$$

Conversely, if

$$C^*C = \lambda I,$$

then all singular values of C equal $\sqrt{\lambda}$, and therefore

$$\kappa_2(C) = 1.$$

□

Computing C^*C gives

$$C^*C = \begin{pmatrix} a^2(1+u^2+v^2) & abv(u+w) \\ abv(u+w) & b^2(1+w^2+v^2) \end{pmatrix}.$$

For $\kappa_2(C) = 1$, the off-diagonal terms must vanish:

$$abv(u+w) = 0.$$

Since $ab \neq 0$, we have

$$v(u+w) = 0.$$

If $u + w = 0$, then $u = -w$. The equality of the diagonal elements becomes

$$a^2(1+w^2+v^2) = b^2(1+w^2+v^2),$$

which is impossible since $a < b$.

Thus

$$v = 0.$$

This leaves

$$N = \begin{pmatrix} u & 0 \\ 0 & w \end{pmatrix}.$$

Solving

$$a^2(1+u^2) = b^2(1+w^2)$$

gives the symmetric matrices achieving perfect conditioning:

$$\boxed{N = \begin{pmatrix} \pm\sqrt{\dfrac{b^2}{a^2}(1+w^2) - 1} & 0 \\ 0 & w \end{pmatrix}, \qquad w \in \mathbb{R}.}$$

We can achieve perfect conditioning for C by choosing such an N.

There is also a special case when $a = b$. In that case, the condition for perfect conditioning becomes

$$u^2 = w^2, \qquad v(u+w) = 0.$$

Thus either

$$u = -w$$

with arbitrary v, or

$$u = w, \qquad v = 0.$$

In particular, $B = 0$ already gives perfect conditioning when $a = b$.

We can also investigate when diagonal choices of N improve the condition number without necessarily making it equal to one.

Let

$$N = \begin{pmatrix} u & 0 \\ 0 & w \end{pmatrix}.$$

Then

$$C = \begin{pmatrix} a(1+iu) & 0 \\ 0 & b(1+iw) \end{pmatrix}.$$

The condition number of $C = A + iB$ is

$$\kappa_2(C) = \frac{\max\left\{a\sqrt{1+u^2},\, b\sqrt{1+w^2}\right\}}{\min\left\{a\sqrt{1+u^2},\, b\sqrt{1+w^2}\right\}}.$$

Requiring strict improvement over the original condition number $\kappa_2(A) = b/a$, and considering both possible orderings of the singular values, yields

$$\boxed{\left(\frac{a}{b}\right)^4 < \frac{1+w^2}{1+u^2} < 1.}$$

Equality at either endpoint gives $\kappa_2(C) = \kappa_2(A)$, and therefore does not yield a strict improvement.

2.2. **Generalization to any** $A \in GL_2(\mathbb{R})$. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det(A) \neq 0.$$

A QR decomposition gives

$$R = \begin{pmatrix} r & s \\ 0 & t \end{pmatrix}, \qquad rt \neq 0,$$

and

$$Q^T Q = I.$$

For

$$B = QNR,$$

we have

$$C = Q(I + iN)R.$$

Therefore,

$$\kappa_2(C) = \kappa_2((I + iN)R)$$

and

$$\kappa_2(A) = \kappa_2(R).$$

We try

$$N = \begin{pmatrix} x & y \\ y & z \end{pmatrix}, \qquad x, y, z \in \mathbb{R},$$

since it worked for diagonal A.

We have

$$I + iN = \begin{pmatrix} 1+ix & iy \\ iy & 1+iz \end{pmatrix}$$

and

$$M = (I + iN)R = \begin{pmatrix} r(1+ix) & s(1+ix) + ity \\ iry & isy + t(1+iz) \end{pmatrix}.$$

For

$$M = \begin{pmatrix} p & q \\ u & v \end{pmatrix},$$

we have

$$M^* = \begin{pmatrix} \overline{p} & \overline{u} \\ \overline{q} & \overline{v} \end{pmatrix}.$$

We want

$$\kappa_2(M) = 1 \iff M^* M = \lambda I$$

with $\lambda > 0$.

Thus

$$M^*M = \begin{pmatrix} |p|^2 + |u|^2 & \overline{p}q + \overline{u}v \\ \overline{q}p + \overline{v}u & |q|^2 + |v|^2 \end{pmatrix} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}.$$

From $(M^*M)_{12} = 0$:

$$\begin{aligned} r(1 - ix)(s(1 + ix) + ity) \\ - iry(isy + t(1 + iz)) = 0. \end{aligned}$$

Expanding,

$$\begin{aligned} rs(1 + x^2) + irty(1 - ix) + ry^2 s \\ - irty(1 + iz) = 0. \end{aligned}$$

Therefore,

$$rs(1 + x^2 + y^2) + rty(x + z) = 0.$$

Since $r \neq 0$, we obtain

$$\boxed{s(1 + x^2 + y^2) + ty(x + z) = 0.} \tag{1}$$

From equating diagonal terms,

$$\begin{aligned} r^2(1 + x^2 + y^2) = s^2(1 + x^2 + y^2) \\ + t^2(1 + y^2 + z^2) \\ + 2sty(x + z). \end{aligned}$$

Using (1),

$$2sty(x + z) = -2s^2(1 + x^2 + y^2).$$

Substituting,

$$\begin{aligned} r^2(1 + x^2 + y^2) = s^2(1 + x^2 + y^2) \\ + t^2(1 + y^2 + z^2) \\ - 2s^2(1 + x^2 + y^2). \end{aligned}$$

Hence

$$\boxed{(r^2 + s^2)(1 + x^2 + y^2) = t^2(1 + y^2 + z^2).} \tag{2}$$

Let's pick variables to satisfy these equations.
Try

$$x = -\frac{t}{r}, \qquad y = \frac{s}{r}.$$

Then

$$1 + x^2 + y^2 = \frac{r^2 + t^2 + s^2}{r^2}.$$

Substituting into (1),

$$\begin{aligned} s\frac{r^2 + t^2 + s^2}{r^2} + t\frac{s}{r}\left(-\frac{t}{r} + z\right) = 0 \\ s\frac{r^2 + s^2}{r^2} + \frac{st^2}{r^2} - \frac{st^2}{r^2} + \frac{stz}{r} = 0. \end{aligned}$$

Thus

$$s\frac{r^2 + s^2}{r^2} + \frac{stz}{r} = 0.$$

If $s \neq 0$, multiplying by r^2/s gives

$$r^2 + s^2 + rtz = 0.$$

Therefore,

$$\boxed{z = -\frac{r^2 + s^2}{rt}.}$$

If $s = 0$, equation (1) is automatically satisfied by $y = 0$. Equation (2) is then satisfied by

$$x = -\frac{t}{r}, \qquad z = -\frac{r}{t}.$$

We now check (2) for the general case:

$$\begin{aligned} \text{LHS} &= (r^2 + s^2)\frac{r^2 + t^2 + s^2}{r^2} \\ &= \frac{(t^2 + r^2 + s^2)(r^2 + s^2)}{r^2}. \end{aligned}$$

On the other hand,

$$\begin{aligned} \text{RHS} &= t^2\left(1 + \frac{s^2}{r^2} + \frac{(r^2 + s^2)^2}{r^2 t^2}\right) \\ &= \frac{t^2(r^2 + s^2) + (r^2 + s^2)^2}{r^2} \\ &= \frac{(t^2 + r^2 + s^2)(r^2 + s^2)}{r^2} \\ &= \text{LHS}. \end{aligned}$$

The equations are satisfied.

For these choices of x, y, z, we construct N. We showed that M^*M has the form λI, meaning

$$\kappa_2(C) = 1.$$

The exact scale factor is

$$(M^*M)_{11} = r^2 + s^2 + t^2.$$

Thus

$$M^*M = (r^2 + s^2 + t^2)I.$$

Therefore,

$$\boxed{\kappa_2(C) = 1.}$$

3. Perfectly Conditioned C for any $A \in GL_n(\mathbb{R})$

We prove that we can make

$$C = A + iB$$

perfectly conditioned for any invertible

$$A \in \mathbb{R}^{n \times n}.$$

Let

$$C = Q(I + iN)R,$$

where

$$A = QR, \qquad B = QNR.$$

We choose N to be real symmetric.

We need

$$C^*C = \lambda I$$

with $\lambda > 0$.

We have

$$\begin{aligned} C^* &= (Q(I+iN)R)^* \\ &= R^T(\overline{I+iN})^T Q^T \\ &= R^T(I-iN)Q^T, \end{aligned}$$

since $N^T = N$.

Therefore,

$$\begin{aligned} C^*C &= R^T(I-iN)Q^TQ(I+iN)R \\ &= R^T(I-iN)(I+iN)R \\ &= R^T(I+N^2)R. \end{aligned}$$

For

$$C^*C = \lambda I,$$

we require

$$I + N^2 = \lambda R^{-T}R^{-1}.$$

Thus

$$N^2 = \lambda R^{-T}R^{-1} - I.$$

We set

$$M = N^2 = \lambda R^{-T}R^{-1} - I.$$

The matrix M must be positive semidefinite.

Since

$$R^{-T}R^{-1} = (R^{-1})^T R^{-1},$$

it is positive definite.

Let μ_i be the eigenvalues of $R^{-T}R^{-1}$. The eigenvalues of M are

$$\lambda\mu_i - 1.$$

Thus we require

$$\lambda\mu_{\min} - 1 \geq 0,$$

which gives

$$\lambda \geq \frac{1}{\mu_{\min}}.$$

Since M is symmetric positive semidefinite, it can be diagonalized as

$$M = U\Lambda U^T$$

by an orthogonal matrix U.

We construct N using the positive semidefinite matrix square root:

$$N = U\Lambda^{1/2}U^T.$$

This satisfies

$$N^2 = M.$$

Therefore,

$$C^*C = R^T(I+N^2)R = \lambda I.$$

Consequently,

$$\kappa_2(C) = \frac{\sigma_{\max}(C)}{\sigma_{\min}(C)} = \frac{\sqrt{\lambda}}{\sqrt{\lambda}} = 1.$$

Thus C is perfectly conditioned.

The condition

$$\lambda \geq \frac{1}{\mu_{\min}(R^{-T}R^{-1})}$$

ensures that M is positive semidefinite.

Since the smallest eigenvalue of $R^{-T}R^{-1}$ is

$$\frac{1}{\sigma_{\max}(R)^2},$$

we may equivalently choose

$$\boxed{\lambda \geq \sigma_{\max}(A)^2.}$$

Since A and R have the same singular values because $A^TA = R^TR$

3.1. **Construction Cost.** Given

$$N^2 = M = aR^{-T}R^{-1} - I$$

with

$$a \geq \sigma_{\max}(R)^2 = \sigma_{\max}(A)^2,$$

how costly is this construction of B?

The eigenvalues of $(RR^T)^{-1}$ are

$$\frac{1}{\sigma_i(R)^2}.$$

Thus the eigenvalues of M are

$$\lambda_i(M) = \frac{a}{\sigma_i(R)^2} - 1.$$

For $M \geq 0$, we need

$$a \geq \sigma_{\max}(R)^2 = \|R\|_2^2.$$

Since

$$B = QNR$$

and $\|Q\|_2 = 1$, we have

$$\|B\|_2 = \|NR\|_2.$$

Now

$$\begin{aligned}
\|B\|_2^2 &= \|B^TB\|_2 \\
&= \|R^TN^2R\|_2 \\
&= \|R^TMR\|_2 \\
&= \|R^T(aR^{-T}R^{-1} - I)R\|_2 \\
&= \|aI - R^TR\|_2.
\end{aligned}$$

The eigenvalues of R^TR are $\sigma_i(A)^2$, so the eigenvalues of B^TB are

$$a - \sigma_i(A)^2.$$

Thus

$$\|B\|_2^2 = \max_i(a - \sigma_i(A)^2) = a - \sigma_{\min}(A)^2.$$

Therefore,

$$\boxed{\|B\|_2 = \sqrt{a - \sigma_{\min}(A)^2}.}$$

For the minimal valid choice

$$a_{\min} = \sigma_{\max}(A)^2,$$

we get

$$\|B\|_{2,\min} = \sqrt{\sigma_{\max}(A)^2 - \sigma_{\min}(A)^2}.$$

Equivalently,

$$\boxed{\|B\|_{2,\min} = \sigma_{\max}(A)\sqrt{1 - \frac{1}{\kappa_2(A)^2}}.}$$

As

$$\kappa_2(A) \to \infty,$$

we have

$$\|B\|_{2,\min} \to \sigma_{\max}(A),$$

provided $\sigma_{\max}(A)$ is fixed.

There is a trade-off:

$$\|B\|_2 = \sqrt{a - \sigma_{\min}(A)^2}$$

increases with a, while

$$\kappa_2(B) = \sqrt{\frac{a - \sigma_{\min}(A)^2}{a - \sigma_{\max}(A)^2}}$$

decreases toward 1 as $a \to \infty$.

The expression for $\kappa_2(B)$ applies when

$$a > \sigma_{\max}(A)^2$$

and $\sigma_{\max}(A) > \sigma_{\min}(A)$.

4. Minimum-Norm Imaginary Perturbation

We can also investigate the trade-off between the norm of B and the condition number of B within the preceding construction.

Assume

$$s = \sigma_{\max}(A) > m = \sigma_{\min}(A) > 0$$

and $a > s^2$.

We have

$$\kappa_2(B)^2 = \frac{a - m^2}{a - s^2}.$$

To impose

$$\kappa_2(B) \le K, \qquad K > 1,$$

we require

$$\frac{a - m^2}{a - s^2} \le K^2.$$

Rearranging,

$$a - m^2 \le K^2 a - K^2 s^2.$$

Therefore,

$$(K^2 - 1)a \ge K^2 s^2 - m^2.$$

Hence

$$\boxed{a \ge \frac{K^2 s^2 - m^2}{K^2 - 1}.}$$

This yields

$$\|B\|_{2,\min}^2 = a - m^2 = \frac{K^2(s^2 - m^2)}{K^2 - 1}.$$

Thus

$$\boxed{\|B\|_{2,\min} = K\sqrt{\frac{s^2 - m^2}{K^2 - 1}}.}$$

This is a trade-off within the particular QR construction family. It is distinct from the general optimization problem considered later, where the constraint is imposed on $\kappa_2(A+iB)$ rather than on $\kappa_2(B)$.

Theorem 4.1 (Minimum-norm real imaginary part for perfect conditioning). *Let $A \in GL_n(\mathbb{R})$ be invertible, with singular values*

$$s = \sigma_{\max}(A), \qquad m = \sigma_{\min}(A) > 0.$$

Then

$$\boxed{\min_{\kappa_2(A+iB)=1} \|B\|_2 = \sqrt{s^2 - m^2}}$$

for $B \in \mathbb{R}^{n\times n}$.

Moreover, this minimum is attained by:

(1) ***SVD construction:*** *If*

$$A = U\Sigma V^T,$$

then

$$B_{\mathrm{SVD}} = U(s^2 I - \Sigma^2)^{1/2} V^T.$$

(2) ***QR construction:*** *If*

$$A = QR,$$

then

$$M = s^2 R^{-T} R^{-1} - I, \qquad N = M^{1/2},$$

and

$$B_{\mathrm{QR}} = QNR.$$

Both constructions satisfy

$$B_{\mathrm{QR}} = B_{\mathrm{SVD}}.$$

Proof. For

$$\kappa_2(A + iB) = 1,$$

all singular values of

$$C = A + iB$$

must be equal. Thus

$$C^*C = \lambda I.$$

Expanding,

$$\begin{aligned} C^*C &= (A^T - iB^T)(A + iB) \\ &= A^TA + B^TB + i(A^TB - B^TA). \end{aligned}$$

Since A, B are real, equating real and imaginary parts gives

$$A^TA + B^TB = \lambda I$$

and

$$A^TB - B^TA = 0.$$

From

$$B^TB = \lambda I - A^TA \geq 0,$$

we know that

$$\lambda \geq s^2.$$

The largest eigenvalue of B^TB is

$$\lambda - m^2.$$

Therefore,

$$\|B\|_2^2 \geq s^2 - m^2.$$

Hence

$$\boxed{\|B\|_2 \geq \sqrt{s^2 - m^2}.}$$

This establishes a global lower bound.
For the SVD construction, let

$$D = (s^2 I - \Sigma^2)^{1/2}.$$

Then

$$B_{\text{SVD}} = UDV^T.$$

We have

$$A^T B_{\text{SVD}} = V\Sigma DV^T$$

and

$$B_{\text{SVD}}^T A = VD\Sigma V^T.$$

Since Σ and D are diagonal, they commute. Thus

$$A^T B_{\text{SVD}} = B_{\text{SVD}}^T A.$$

Furthermore,

$$\begin{aligned} C^*C &= A^T A + B_{\text{SVD}}^T B_{\text{SVD}} \\ &= V\Sigma^2 V^T + VD^2 V^T \\ &= V(\Sigma^2 + D^2)V^T \\ &= Vs^2 IV^T \\ &= s^2 I. \end{aligned}$$

Thus

$$\kappa_2(C) = 1.$$

Also,

$$\|B_{\text{SVD}}\|_2 = \|D\|_2 = \sqrt{s^2 - m^2}.$$

Therefore, the lower bound is attained.
For the QR construction, let

$$A = QR.$$

Since

$$A = U\Sigma V^T,$$

we have

$$R = Q^T U\Sigma V^T.$$

Define

$$W = Q^T U.$$

Since Q and U are orthogonal, W is orthogonal. Hence

$$R = W\Sigma V^T.$$

It follows that

$$R^{-T}R^{-1} = W\Sigma^{-2}W^T.$$

Therefore,

$$M = s^2 R^{-T}R^{-1} - I = W(s^2\Sigma^{-2} - I)W^T.$$

Taking the positive semidefinite square root,

$$N = W(s^2\Sigma^{-2} - I)^{1/2}W^T.$$

Substituting into B_{QR} gives

$$\begin{aligned} B_{\mathrm{QR}} &= QNR \\ &= QW(s^2\Sigma^{-2} - I)^{1/2} W^T W \Sigma V^T \\ &= QW(s^2\Sigma^{-2} - I)^{1/2} \Sigma V^T. \end{aligned}$$

Since $QW = U$, we obtain

$$B_{\mathrm{QR}} = U(s^2\Sigma^{-2} - I)^{1/2}\Sigma V^T.$$

Because the matrices involved are diagonal,

$$(s^2\Sigma^{-2} - I)^{1/2}\Sigma = (s^2 I - \Sigma^2)^{1/2}.$$

Consequently,

$$B_{\mathrm{QR}} = U(s^2 I - \Sigma^2)^{1/2} V^T = B_{\mathrm{SVD}}.$$

Thus the QR and SVD constructions give exactly the same matrix, and both attain the global lower bound. □

5. Iterative Method for $Ax = b$

We proved that for C to be perfectly conditioned while minimizing $\|B\|_2$, we have

$$A^T A + B^T B = s^2 I,$$

where

$$s = \sigma_{\max}(A).$$

For the system

$$Ax = b,$$

multiplying by A^T gives

$$A^T A x = A^T b.$$

Using

$$A^T A = s^2 I - B^T B,$$

we obtain

$$\begin{aligned} (s^2 I - B^T B)x &= A^T b, \\ s^2 x &= A^T b + B^T B x. \end{aligned}$$

This suggests the fixed-point iteration

$$\boxed{x_{k+1} = \frac{1}{s^2} A^T b + \frac{1}{s^2} B^T B x_k, \qquad k \geq 0.}$$

Substituting

$$B^T B = s^2 I - A^T A,$$

we get

$$\begin{aligned} x_{k+1} &= \frac{1}{s^2} A^T b + \frac{1}{s^2}(s^2 I - A^T A)x_k \\ &= x_k + \frac{1}{s^2} A^T (b - A x_k). \end{aligned}$$

Thus

$$\boxed{x_{k+1} = x_k + \frac{1}{s^2} A^T (b - A x_k).}$$

This is the classical Landweber iteration, which is also gradient descent applied to the least-squares objective

$$f(x) = \frac{1}{2}\|Ax - b\|_2^2.$$

Geometrically, the iteration is equivalent to extracting the real solution from the structure of the minimum-norm perfectly conditioned complex counterpart.

Theorem 5.1 (Fixed-point induced by the minimum-norm construction). *Let* $A \in GL_n(\mathbb{R})$, *with singular values*

$$s = \sigma_{\max}(A), \qquad m = \sigma_{\min}(A) > 0.$$

Let $B \in \mathbb{R}^{n\times n}$ *be the minimum-norm matrix from the preceding theorem.*
For any $b \in \mathbb{R}^n$, *consider*

$$x_{k+1} = \frac{A^T b}{s^2} + \frac{B^T B}{s^2} x_k.$$

Then:

i) ***Convergence:*** *For any initial vector* $x_0 \in \mathbb{R}^n$, *the sequence* (x_k) *converges to the unique solution*

$$x_* = A^{-1} b.$$

ii) ***Convergence rate:*** *For all* $k \geq 0$,

$$\|x_{k+1} - x_*\|_2 \leq \left(1 - \frac{m^2}{s^2}\right) \|x_k - x_*\|_2,$$

with contraction factor

$$q = 1 - \frac{1}{\kappa_2(A)^2} < 1.$$

Proof. Let

$$c = \frac{A^T b}{s^2}, \qquad T = \frac{B^T B}{s^2}.$$

We have

$$x_{k+1} = T x_k + c.$$

The exact solution $x_* = A^{-1} b$ is a fixed point, since

$$T x_* + c = \frac{B^T B}{s^2} x_* + \frac{A^T b}{s^2}.$$

Using

$$B^T B = s^2 I - A^T A,$$

we obtain

$$\begin{aligned} T x_* + c &= \frac{s^2 I - A^T A}{s^2} x_* + \frac{A^T b}{s^2} \\ &= x_* - \frac{A^T A x_*}{s^2} + \frac{A^T b}{s^2}. \end{aligned}$$

Since $A x_* = b$, we have

$$A^T A x_* = A^T b.$$

Thus

$$T x_* + c = x_*.$$

The error

$$e_k = x_k - x_*$$

satisfies

$$e_{k+1} = T e_k.$$

Therefore,

$$e_k = T^k e_0.$$

The operator is

$$T = \frac{B^T B}{s^2} = I - \frac{A^T A}{s^2}.$$

Its eigenvalues are

$$\lambda_i(T) = 1 - \frac{\sigma_i(A)^2}{s^2}.$$

Since

$$m \le \sigma_i(A) \le s,$$

we have

$$0 \le \lambda_i(T) \le 1 - \frac{m^2}{s^2}.$$

In particular,

$$\lambda_{\max}(T) = 1 - \frac{m^2}{s^2} < 1.$$

Since T is symmetric positive semidefinite, its spectral norm equals its largest eigenvalue:

$$\|T\|_2 = \lambda_{\max}(T) = 1 - \frac{m^2}{s^2} < 1.$$

Hence

$$\|e_{k+1}\|_2 = \|Te_k\|_2 \le \|T\|_2 \|e_k\|_2.$$

Thus

$$\|e_{k+1}\|_2 \le \left(1 - \frac{m^2}{s^2}\right) \|e_k\|_2.$$

Iterating this inequality,

$$\|e_k\|_2 \le \left(1 - \frac{m^2}{s^2}\right)^k \|e_0\|_2.$$

Since

$$0 \le 1 - \frac{m^2}{s^2} < 1,$$

we have

$$e_k \to 0$$

as $k \to \infty$.

Therefore,

$$x_k \to x_* = A^{-1}b.$$

Finally,

$$1 - \frac{m^2}{s^2} = 1 - \frac{1}{\kappa_2(A)^2}.$$

This proves both convergence and the stated convergence rate. □

6. Generalized Bounds

Theorem 6.1 (Minimum-norm perturbation under a prescribed condition number bound). *Let $A \in GL_n(\mathbb{R})$ with*

$$s = \sigma_{\max}(A), \qquad m = \sigma_{\min}(A) > 0.$$

For any $K \ge 1$, consider the optimization problem

$$\min_{B \in \mathbb{R}^{n \times n}} \|B\|_2$$

subject to

$$\kappa_2(A + iB) \le K.$$

The minimum is

$$\boxed{\|B\|_{2,\min} = \sqrt{\max\left\{0, \frac{s^2}{K^2} - m^2\right\}}.}$$

Let

$$A = U\Sigma V^T$$

be a singular value decomposition of A, where U, V are orthogonal and

$$\Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_n), \qquad s = \sigma_1 \geq \cdots \geq \sigma_n = m > 0.$$

Define

$$\tau = \frac{s}{K}$$

and

$$D = \left(\tau^2 I - \Sigma^2\right)_+^{1/2}.$$

Then the matrix

$$\boxed{B_{\mathrm{SVD}} = UDV^T}$$

attains the minimum.

Here $(\cdot)_+$ denotes the positive part of a real symmetric matrix: if

$$H = W \operatorname{diag}(\lambda_1, \dots, \lambda_n) W^T,$$

where W is orthogonal, then

$$H_+ = W \operatorname{diag}(\max\{\lambda_1, 0\}, \dots, \max\{\lambda_n, 0\}) W^T.$$

Alternatively, let

$$A = QR$$

be a QR decomposition of A, where Q is orthogonal and R is invertible and upper triangular. Define

$$P = \left(\tau^2 I - R^T R\right)_+.$$

Then the QR construction

$$\boxed{B_{\mathrm{QR}} = Q(R^{-T} P R^{-1})^{1/2} R}$$

also attains the minimum.

Moreover, the two constructions produce exactly the same matrix:

$$\boxed{B_{\mathrm{QR}} = B_{\mathrm{SVD}}.}$$

In both cases,

$$\boxed{\kappa_2(A + iB_*) = \frac{s}{\max\{m, s/K\}} \leq K.}$$

Proof. We first establish a lower bound for the norm of any feasible matrix B. We then construct a matrix attaining this bound using the singular value decomposition. Finally, we derive the QR construction and prove that it is identical to the SVD construction.

1. Lower bound.

Let $B \in \mathbb{R}^{n \times n}$ be arbitrary and set

$$C = A + iB.$$

Let

$$t = \|B\|_2.$$

Since A and B are real matrices, for every real unit vector v we have

$$\|(A + iB)v\|_2^2 = \|Av\|_2^2 + \|Bv\|_2^2.$$

Indeed, the real and imaginary parts of $(A + iB)v$ are Av and Bv, respectively, and hence their squared norms add.

Let v_s and v_m be real unit right singular vectors of A corresponding to its largest and smallest singular values, respectively. Thus,

$$\|Av_s\|_2 = s, \qquad \|Av_m\|_2 = m.$$

Evaluating C at v_s, we obtain

$$\begin{aligned} \|Cv_s\|_2^2 &= \|Av_s\|_2^2 + \|Bv_s\|_2^2 \\ &= s^2 + \|Bv_s\|_2^2 \\ &\geq s^2. \end{aligned}$$

Since the largest singular value is the maximum of $\|Cv\|_2$ over all unit vectors,

$$\sigma_{\max}(C)^2 \geq s^2.$$

Similarly, evaluating C at v_m gives

$$\begin{aligned} \|Cv_m\|_2^2 &= \|Av_m\|_2^2 + \|Bv_m\|_2^2 \\ &= m^2 + \|Bv_m\|_2^2 \\ &\leq m^2 + t^2. \end{aligned}$$

Since the smallest singular value is the minimum of $\|Cv\|_2$ over all unit vectors,

$$\sigma_{\min}(C)^2 \leq m^2 + t^2.$$

Consequently,

$$\begin{aligned} \kappa_2(C)^2 &= \frac{\sigma_{\max}(C)^2}{\sigma_{\min}(C)^2} \\ &\geq \frac{s^2}{m^2 + t^2}. \end{aligned}$$

For B to satisfy the prescribed condition number bound, we must have

$$\kappa_2(C) \leq K.$$

Therefore,

$$K^2 \geq \frac{s^2}{m^2 + t^2}.$$

Rearranging,

$$m^2 + t^2 \geq \frac{s^2}{K^2},$$

and hence

$$t^2 \geq \frac{s^2}{K^2} - m^2.$$

Since $t^2 \geq 0$, we obtain the global lower bound

$$\boxed{\|B\|_2 \geq \sqrt{\max\left\{0, \frac{s^2}{K^2} - m^2\right\}}.}$$

We now construct a matrix attaining this bound.

2. Construction using the singular value decomposition.
Let

$$A = U\Sigma V^T$$

be a singular value decomposition of A, where U, V are orthogonal and

$$\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_n).$$

Recall that

$$s = \sigma_1, \qquad m = \sigma_n.$$

Define the threshold

$$\tau = \frac{s}{K}.$$

Our objective is to construct a perturbation that raises every singular value of A smaller than τ to τ, while leaving all the other singular values unchanged.

To achieve this, define the diagonal matrix

$$D = \left(\tau^2 I - \Sigma^2\right)_+^{1/2}.$$

More explicitly,

$$D = \operatorname{diag}(d_1, \ldots, d_n),$$

where

$$d_j = \sqrt{\max\{0, \tau^2 - \sigma_j^2\}}.$$

Thus, if $\sigma_j \geq \tau$, then $d_j = 0$, whereas if $\sigma_j < \tau$, then

$$d_j = \sqrt{\tau^2 - \sigma_j^2}.$$

We now set

$$\boxed{B_{\mathrm{SVD}} = UDV^T.}$$

Consider

$$C = A + iB_{\mathrm{SVD}}.$$

Using the SVD of A, we obtain

$$\begin{aligned} C &= U\Sigma V^T + iUDV^T \\ &= U(\Sigma + iD)V^T. \end{aligned}$$

Since U and V are orthogonal, multiplication by these matrices does not change singular values. Therefore, the singular values of C are those of $\Sigma + iD$.

Now,

$$\begin{aligned} (\Sigma + iD)^*(\Sigma + iD) &= (\Sigma - iD)(\Sigma + iD) \\ &= \Sigma^2 + D^2. \end{aligned}$$

The last equality follows because Σ and D are diagonal matrices and therefore commute.

By the definition of D,

$$D^2 = (\tau^2 I - \Sigma^2)_+.$$

Consequently,

$$\Sigma^2 + D^2 = \operatorname{diag}\left(\max\{\sigma_1^2, \tau^2\}, \ldots, \max\{\sigma_n^2, \tau^2\}\right).$$

It follows that the singular values of C are precisely

$$\boxed{\sigma_j(C) = \max\{\sigma_j(A), \tau\}, \qquad j = 1, \ldots, n.}$$

In particular, since $\tau = s/K \leq s$,

$$\sigma_{\max}(C) = s,$$

while

$$\sigma_{\min}(C) = \max\{m, \tau\}.$$

Therefore,

$$\boxed{\kappa_2(C) = \frac{s}{\max\{m, \tau\}} = \frac{s}{\max\{m, s/K\}} \leq K.}$$

It remains to calculate the norm of the constructed perturbation.

Since U and V are orthogonal,

$$\|B_{\mathrm{SVD}}\|_2 = \|UDV^T\|_2 = \|D\|_2.$$

As D is diagonal and positive semidefinite,

$$\|D\|_2 = \max_{1\le j\le n} d_j.$$

The largest value occurs at the smallest singular value m. Hence,

$$\begin{aligned}\|B_{\mathrm{SVD}}\|_2 &= \sqrt{\max\{0, \tau^2 - m^2\}} \\ &= \sqrt{\max\left\{0, \frac{s^2}{K^2} - m^2\right\}}.\end{aligned}$$

Thus, the lower bound is attained, and the SVD construction is optimal.

3. Construction using the QR decomposition.

We now derive an alternative expression for the same optimal perturbation using the QR decomposition.

Let

$$A = QR,$$

where Q is orthogonal and R is invertible and upper triangular.

We seek a perturbation of the form

$$B = QNR,$$

where N is real symmetric and positive semidefinite.

This form is convenient because

$$A + iB = QR + iQNR = Q(I + iN)R.$$

Furthermore,

$$\begin{aligned}B^TB &= (QNR)^T(QNR) \\ &= R^TN^TQ^TQNR \\ &= R^TN^2R,\end{aligned}$$

where we have used $Q^TQ = I$ and $N^T = N$.

Recall that

$$A^TA = R^TR.$$

We want the eigenvalues of

$$A^TA + B^TB$$

to be

$$\max\{\sigma_j(A)^2, \tau^2\}, \qquad j = 1, \ldots, n.$$

This suggests defining

$$\boxed{P = (\tau^2 I - R^TR)_+.}$$

Indeed, if we can choose N such that

$$B^TB = P,$$

then

$$A^TA + B^TB = R^TR + P$$

will have exactly the desired eigenvalues.

Since

$$B^TB = R^TN^2R,$$

we require

$$R^TN^2R = P.$$

Multiplying from the left by R^{-T} and from the right by R^{-1} gives

$$N^2 = R^{-T}PR^{-1}.$$

The matrix on the right is symmetric positive semidefinite, because P is symmetric positive semidefinite. We can therefore define N as its principal symmetric positive semidefinite square root:

$$\boxed{N = (R^{-T}PR^{-1})^{1/2}.}$$

Our QR construction is consequently

$$\boxed{B_{\mathrm{QR}} = Q(R^{-T}PR^{-1})^{1/2}R.}$$

We verify that this construction has the required properties.
By definition,

$$N^2 = R^{-T}PR^{-1}.$$

Hence,

$$\begin{aligned} B_{\mathrm{QR}}^T B_{\mathrm{QR}} &= R^T N^2 R \\ &= R^T R^{-T} P R^{-1} R \\ &= P. \end{aligned}$$

Furthermore,

$$A^T B_{\mathrm{QR}} = R^T N R.$$

Since N is symmetric,

$$(R^T N R)^T = R^T N R.$$

Therefore,

$$A^T B_{\mathrm{QR}} = B_{\mathrm{QR}}^T A.$$

For $C = A + iB_{\mathrm{QR}}$, we consequently obtain

$$\begin{aligned} C^*C &= (A - iB_{\mathrm{QR}}^T)(A + iB_{\mathrm{QR}}) \\ &= A^T A + B_{\mathrm{QR}}^T B_{\mathrm{QR}} \\ &= R^T R + P. \end{aligned}$$

By the definition of P,

$$R^T R + P = R^T R + (\tau^2 I - R^T R)_+.$$

Thus, the eigenvalues of C^*C are

$$\max\{\sigma_1(A)^2, \tau^2\}, \ldots, \max\{\sigma_n(A)^2, \tau^2\}.$$

It follows that the singular values of C are

$$\sigma_j(C) = \max\{\sigma_j(A), \tau\}.$$

In particular,

$$\kappa_2(C) = \frac{s}{\max\{m, \tau\}} \leq K.$$

Moreover, since

$$B_{\mathrm{QR}}^T B_{\mathrm{QR}} = P,$$

we have

$$\begin{aligned} \|B_{\mathrm{QR}}\|_2^2 &= \|B_{\mathrm{QR}}^T B_{\mathrm{QR}}\|_2 \\ &= \|P\|_2 \\ &= \max\{0, \tau^2 - m^2\}. \end{aligned}$$

Therefore,

$$\boxed{\|B_{\mathrm{QR}}\|_2 = \sqrt{\max\left\{0, \frac{s^2}{K^2} - m^2\right\}}.}$$

Thus, the QR construction also attains the lower bound.

4. Equivalence of the QR and SVD Constructions

Let

$$A = U\Sigma V^T, \qquad A = QR,$$

be the singular value decomposition and the QR decomposition of A, respectively.

Define

$$W = Q^T U.$$

Since Q and U are orthogonal matrices, W is also orthogonal. Moreover,

$$R = Q^T A = W\Sigma V^T.$$

Writing

$$D = (\tau^2 I - \Sigma^2)_+^{1/2},$$

we obtain

$$P = V(\tau^2 I - \Sigma^2)_+ V^T = VD^2 V^T.$$

Consequently,

$$\begin{aligned} R^{-T} P R^{-1} &= (W\Sigma V^T)^{-T} V D^2 V^T (W\Sigma V^T)^{-1} \\ &= W\Sigma^{-1} D^2 \Sigma^{-1} W^T. \end{aligned}$$

Since Σ and D are diagonal matrices, they commute. Hence, the positive-semidefinite square root of $R^{-T}PR^{-1}$ is

$$N = W\Sigma^{-1} D W^T.$$

Therefore, the QR-based construction gives

$$\begin{aligned} B_{\mathrm{QR}} &= QNR \\ &= QW\Sigma^{-1} D W^T W\Sigma V^T \\ &= QWDV^T \\ &= UDV^T \\ &= B_{\mathrm{SVD}}. \end{aligned}$$

Thus, the two constructions yield exactly the same matrix:

$$\boxed{B_{\mathrm{QR}} = B_{\mathrm{SVD}}.}$$

□

References

1. Zhen Dai, Lek-Heng Lim, and Ke Ye, *Complex matrix inversion via real matrix inversions*, 2023.